

PALAEONTOLOGICAL IMPACT ASSESSMENT

Proposed Walmer Housing Development, ERF 11305 in

Walmer, Gqeberha, Nelson Mandela Bay Metropolitan Municipality, Eastern Cape

March 2026

COMPILED FOR:

Abantu Environmental Services (Pty) Ltd

Desktop Study (Phase 1)

Short *Curriculum vitae*:

Mr Silindokuhle S.S. Mavuso recently completed his PhD in the School of Geosciences at the University of the Witwatersrand. He holds a BSc degree in Geology, Archaeology, and Environmental and Geographical Sciences from the University of Cape Town, as well as a BSc Honours in Geology and Palaeontology and an MSc in Geology from the University of the Witwatersrand. During his postgraduate training he was also an Erasmus Mundus Masters mobility scholar at Paul Sabatier III University in Toulouse, France.

His research focuses on Plio-Pleistocene palaeogeology, with particular emphasis on reconstructing palaeoenvironments linked to early human evolution. His work combines approaches from palaeontology, palaeoanthropology, sedimentology, stratigraphy, micromorphology, geochemistry, and geochronology to better understand the environmental contexts of fossil-bearing deposits.

He is currently a lecturer in geology and palaeoanthropology at Rhodes University and a research affiliate at Arizona State University. He has participated in research projects at several major fossil localities across eastern and southern Africa and has extensive experience working within the sedimentary successions of the Karoo Basin.

Declaration of Independence

This report has been compiled by Mr. Silindokuhle Mavuso, sub-contracted by Abantu Environmental Services, South Africa. The views expressed in this report are entirely those of the author and no other interest was displayed during the decision making process for the Project.

Legislative requirements: South African Heritage Resources Agency (SAHRA) for issue of permits if necessary.

National Heritage Resources Act (Act No. 25 of 1999). An electronic copy of this report must be supplied to SAHRA

EXECUTIVE SUMMARY

Abantu Environmental Services (Pty) Ltd appointed Mr Silindokuhle S.S. Mavuso to undertake a Desktop Palaeontological Impact Assessment (PIA) for the proposed Walmer Housing Development on ERF 11305 in Walmer, Gqeberha, within the Nelson Mandela Bay Metropolitan Municipality, Eastern Cape. The study was conducted in accordance with the requirements of Appendix 6 of the 2014 Environmental Impact Assessment Regulations under the National Environmental Management Act (Act No. 107 of 1998), as well as the National Heritage Resources Act (Act No. 25 of 1999).

The purpose of the study was to determine the potential impact of the proposed development on fossil heritage resources and to provide recommendations for mitigation where necessary. The assessment was conducted as a desktop study based on available geological maps, published scientific literature, the SAHRIS palaeosensitivity map, satellite imagery, and previously conducted palaeontological studies in the broader region.

The study area is underlain by sediments of the Algoa Group, most likely belonging to the Salnova Formation. These Cenozoic deposits consist mainly of calcareous sandstones, shelly limestones, and associated coastal sediments deposited in shallow marine and coastal environments along the south-eastern Cape coast. Fossil assemblages within these deposits are typically dominated by marine invertebrates such as molluscs, as well as occasional echinoids and other marine organisms. These fossils provide important information about past coastal ecosystems and environmental change along the southern African coastline.

The SAHRIS palaeosensitivity map classifies the geological units underlying the study area as having very high palaeontological sensitivity, reflecting the recognised fossil potential of Algoa Group sediments. However, the development footprint is located within an already urbanised area where natural geological outcrops are absent and the underlying strata are largely covered by soil and anthropogenic deposits. Previous palaeontological work conducted in the Walmer area for nearby infrastructure development reported no fossil material and concluded that the development footprint was of low palaeontological significance.

Based on the available geological and palaeontological information, the probability of encountering fossil material within the development footprint is considered low to moderate, although the presence of isolated fossil material within subsurface deposits cannot be completely excluded. It is therefore recommended that the proposed development may proceed, provided that a Chance Find Protocol is implemented during excavation activities to ensure that any fossil material that may be encountered is appropriately managed in accordance with the National Heritage Resources Act.

1. INTRODUCTION

Abantu Environmental Services (PTY) Ltd appointed Mr Silindokuhle Mavuso, an independent specialist in palaeontology and geology, to undertake a Desktop Palaeontological Impact Assessment for the proposed Walmer Housing Development on ERF 11305 in Walmer, Gqeberha, within the Nelson Mandela Bay Metropolitan Municipality, Eastern Cape. This report has been prepared in accordance with the requirements set out in Appendix 6 of the 2014 Environmental Impact Assessment Regulations, promulgated under Section 24(5) of the National Environmental Management Act (Act No. 107 of 1998). These regulations require specialist input where heritage resources, including palaeontological heritage, may potentially be impacted by development activities.

South Africa possesses an internationally significant fossil record that provides critical insights into the evolution of life on Earth. Fossil deposits from the country document key stages in the history of life, ranging from early marine organisms and the emergence of terrestrial vertebrates to the evolutionary origins of dinosaurs, mammals, and ultimately humans. In addition to these major scientific contributions, ongoing discoveries continue to refine our understanding of ancient ecosystems and the environmental conditions that shaped them. These records are particularly important for reconstructing the geological and ecological history of Gondwana and its constituent landmasses. For this reason, the preservation and responsible management of fossil-bearing deposits are of national and international importance.

In terms of the National Heritage Resources Act (Act No. 25 of 1999), fossils and fossil-bearing geological contexts are protected heritage resources and may not be damaged, disturbed, or destroyed without appropriate authorisation. Palaeontological Impact Assessments are therefore undertaken to evaluate whether proposed development activities may affect fossil heritage and to recommend appropriate mitigation measures where necessary.

This desktop study evaluates the likelihood that fossil material may occur within the geological units underlying the study area and assesses the potential impact of the proposed development

on such resources. The study area is located within the suburb of Walmer in the city of Gqeberha, Eastern Cape Province, and forms part of ERF 11305 where the proposed housing development is planned (Figure 1).



Figure 1. Satellite image showing the location of the proposed development (yellow) in relation to the broader geographical area.

1.1. LEGISLATION

National Heritage Resources Act (25 of 1999)

Cultural Heritage in South Africa, includes all heritage resources, is protected by the National Heritage Resources Act (Act No. 25 of 1999) (NHRA). Heritage resources as defined in Section 3 of the Act include *“all objects recovered from the soil or waters of South Africa, including archaeological and palaeontological objects and material, meteorites and rare geological specimens”*.

The identification, evaluation and assessment of any cultural heritage site, artefact or finds in the South African context is required and governed by the following legislation:

- National Environmental Management Act (NEMA) Act No. 107 of 1998

- National Heritage Resources Act (NHRA) Act No. 25 of 1999
- Minerals and Petroleum Resources Development Act (MPRDA) Act No. 28 of 2002
- Notice 648 of the Government Gazette 45421- general requirements for undertaking an initial site sensitivity verification where no specific assessment protocol has been identified.

The next section in each Act is directly applicable to the identification, assessment, and evaluation of cultural heritage resources.

GNR 982 (Government Gazette 38282, 14 December 2014) promulgated under the National Environmental Management Act (NEMA) Act No. 107 of 1998

- Basic Assessment Report (BAR) – Regulations 19 and 23
- Environmental Impacts Assessment (EIA) – Regulation 23
- Environmental Scoping Report (ESR) – Regulation 21
- Environmental Management Programme (EMPr) – Regulations 19 and 23

National Heritage Resources Act (NHRA) Act No. 25 of 1999

- Protection of Heritage Resources – Sections 34 to 36
- Heritage Resources Management – Section 38

The NEMA (No. 107 of 1998) states that an integrated EMP should (23:2 (b)) “...identify, predict and evaluate the actual and potential impact on the environment, socio-economic conditions and cultural heritage”

In agreement with legislative requirements, EIA rating standards as well as SAHRA policies a comprehensive and legally compatible PIA report has been compiled.

Palaeontological heritage is exceptional and non-renewable and is protected by the NHRA. Palaeontological resources and may not be unearthed, broken, moved, or destroyed by any development without prior assessment and without a permit from the relevant heritage resources authority as per section 35 of the NHRA.

This Palaeontological Impact assessment forms part of the Heritage Impact Assessment (HIA) and adheres to the conditions of the Act. According to Section 38(1), an HIA is required to assess any potential impacts to palaeontological heritage within the development footprint where:

- the construction of a road, wall, power line, pipeline, canal or other similar form of linear development or barrier exceeding 300 m in length.
- the construction of a bridge or similar structure exceeding 50 m in length.
- any development or other activity which will change the character of a site—
 - exceeding 5 000 m² in extent; or
 - involving three or more existing erven or subdivisions thereof; or
 - involving three or more erven or divisions thereof which have been consolidated within the past five years; or
 - the costs of which will exceed a sum set in terms of regulations by SAHRA or a provincial heritage resources authority or
 - the re-zoning of a site exceeding 10 000 m² in extent or

any other category of development provided for in regulations by SAHRA or a Provincial heritage resources authority.

METHODOLOGY AND TERMS OF REFERENCE

Terms of reference: Mr. S.S. Mavuso is a palaeogeologist commissioned to conduct a palaeontological impact assessment: desktop study to ascertain if any palaeontological sensitive material is present in the development area. This study will advise on the impact on fossil heritage mitigation or conservation necessary, if any.

This Palaeontological Impact assessment assesses the development's potential impact on the fossil heritage. This Palaeontological Assessment is part of the HIA Report.

The PIA's goals are to:

- 1) identify the palaeontological significance of the rock formations in the footprint;
- 2) evaluate the palaeontological magnitude of the formations;
- 3) clarify the impact on fossil heritage; and
- 4) make recommendations for how the developer might protect and minimize potential harm to fossil heritage, according to the "SAHRA APM Guidelines: Minimum Standards for the Archaeological and Palaeontological Components of Impact Assessment Reports".

Calculations of the palaeontological state of each stratigraphic and lithological segment and the potential impact of development on fossil history consider the palaeontological status of the rocks, the type of development, and the amount of bedrock removed.

The preparation of the palaeontological desktop study will be mostly based on assessing the geology of the area to assess for rocks that may house potentially fossiliferous rock units. This will be largely based on geological maps looking at formations and groups etc. The Provisional DFFE Screening Tool, the SAHRIS Palaeosensitivity map, all Palaeontological Impact Assessment reports for the same area, Google Earth images, topographical and geological maps, as well as academic articles about specimens from the development area and Assemblage Zones, are all used to create desktop Palaeontological Impact Assessments. This will include a plot of nearby collected fossil data in relation to the site being assessed. The range of this activity will be done at a 100 km radius. Should it produce abundant fossil, particularly near the development site, the scale of the scope will go into more recent literature as well as procuring more fossil databases, particularly from museums. As most geological data was done decades ago by the Council of Geoscience and the South African Committee for Stratigraphy, a research study shall also be conducted for updated information by both published literature as well as senior postgraduate work procured from all the schools of geology and palaeontology in the country.

2. GEOLOGICAL AND PALAEONTOLOGICAL HISTORY

The proposed development site in the Cenozoic rocks of Algoa Group (Figure 2).

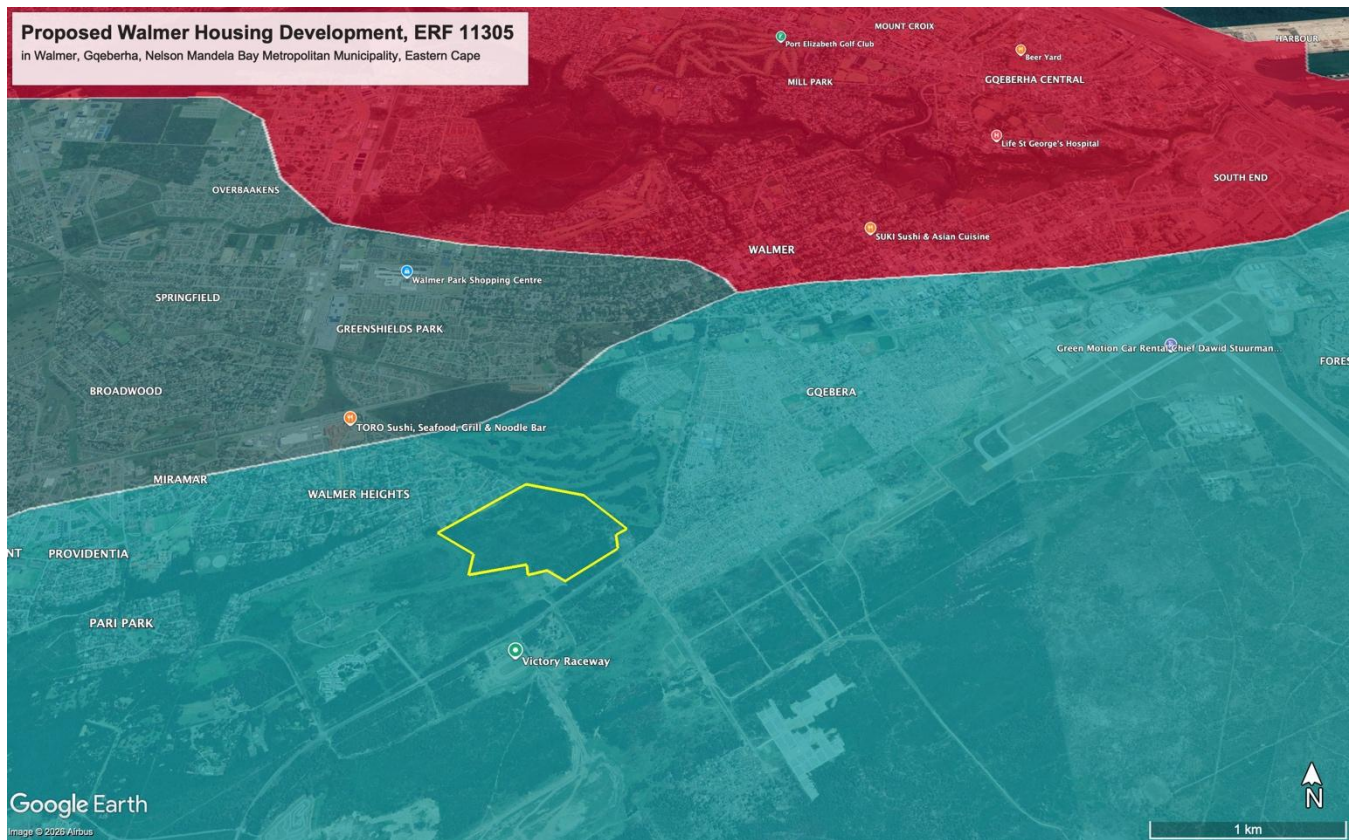


Figure 2. Map showing study area in yellow polygon (digital .kml adapted from the 1:250 000 Geology Map of 3324 Port Elizabeth ; Council of Geosciences).

Legend colour	Formation	Lithology	Age
	Salnova	Calcareous sandstone, shelly limestone, conglomerate	Quaternary
	Salnova	Greenish grey mudstone, sandstone	Quaternary
	Nanaga	Reddish and greenish mudstone, sandstone	Tertiary

Although the digital .kml geological map of South Africa (Council of Geosciences) represents undifferentiated parts of the Algoa Group, the Port Elizabeth 3324 1: 250 000 Geology Map suggests that it was the Salnova Formation (Fourie, 2010).

2.1. THE ALGOA GROUP

The Algoa Group consists of a succession of sedimentary formations distributed along the south-eastern coastline of the Eastern Cape, South Africa. These deposits are of Cenozoic age and are largely associated with marine and coastal depositional systems, with significant aeolian influences (Le Roux, 1989). The group forms an important component of the coastal stratigraphy of the region, recording the interaction between sea-level changes, coastal processes, and sediment accumulation throughout the Cenozoic (Shone and Paszkowski, 1995).

Lithologically, the Algoa Group is characterised by calcareous clastic sediments. Typical rock types include calcareous sandstones, sandy limestones, conglomerates, and coquinite deposits. These lithologies reflect sedimentation within dynamic nearshore and coastal environments where marine and aeolian processes interacted (Toerien and Hill, 1989). The deposits commonly display features indicative of high-energy depositional conditions, including beach ridge systems, coastal dune complexes, and shallow marine facies. These environments were strongly influenced by wave action, wind reworking, and coastal currents that shaped sediment transport and accumulation along the shoreline (Roberts et al., 2006).

The stratigraphic succession of the Algoa Group provides important insights into the palaeoenvironmental history of the south-eastern Cape during the Cenozoic. The preserved sedimentary record reflects patterns of sediment supply and depositional processes and contributes to broader interpretations of regional sea-level fluctuations, coastal landscape evolution, and climatic influences through time (Le Roux, 1989).

2.1.1. THE SALNOVA FORMATION

The Salnova Formation forms part of the Algoa Group, a succession of Cenozoic coastal and shallow marine deposits preserved along the south-eastern coastline of South Africa within the Algoa Basin (Le Roux, 1989; Shone and Paszkowski, 1995). These sediments were deposited in response to fluctuating sea levels and evolving coastal dynamics during the Neogene to Quaternary, recording the development of marine, shoreline, and aeolian depositional systems along the margin of the Eastern Cape. The formation therefore represents an important component of the coastal stratigraphy of the Nelson Mandela Bay region.

Lithologically, the Salnova Formation is composed predominantly of calcareous sands and sandstones, with locally developed shell-rich horizons, calcarenites, and coquina layers derived from marine bioclastic material. These sediments are commonly associated with coastal dune systems, beach ridge complexes, and shallow marine deposits formed in high-energy coastal environments (Toerien and Hill, 1989; Roberts et al., 2006). Sedimentary structures and facies relationships indicate deposition under conditions strongly influenced by wave action, coastal currents, and aeolian reworking, reflecting the dynamic interaction between marine and terrestrial processes along the coastal margin.

Within the broader context of the Algoa Basin stratigraphy, the Salnova Formation contributes to reconstructions of coastal evolution and palaeoenvironmental change in the south-eastern Cape. Its sedimentary record reflects shifts in shoreline position, sediment supply, and climate-driven sea-level fluctuations that have shaped the modern coastal landscape of the Eastern Cape (Le Roux, 1989; Shone and Paszkowski, 1995).

2.2. PALAEOONTOLOGY

The fossil record associated with the Salnova Formation reflects its origin within coastal and shallow marine depositional environments of the Algoa Basin during the late Cenozoic. Fossil assemblages within the Algoa Group are typically dominated by marine invertebrates,

particularly molluscs such as bivalves and gastropods, which were deposited within nearshore marine settings and subsequently preserved within calcareous sands, calcarenites, and shell-rich coquina horizons (Le Roux, 1989; Roberts et al., 2006). These faunal assemblages are characteristic of temperate coastal marine ecosystems and provide important evidence for reconstructing palaeoenvironmental conditions along the south-eastern Cape coast.

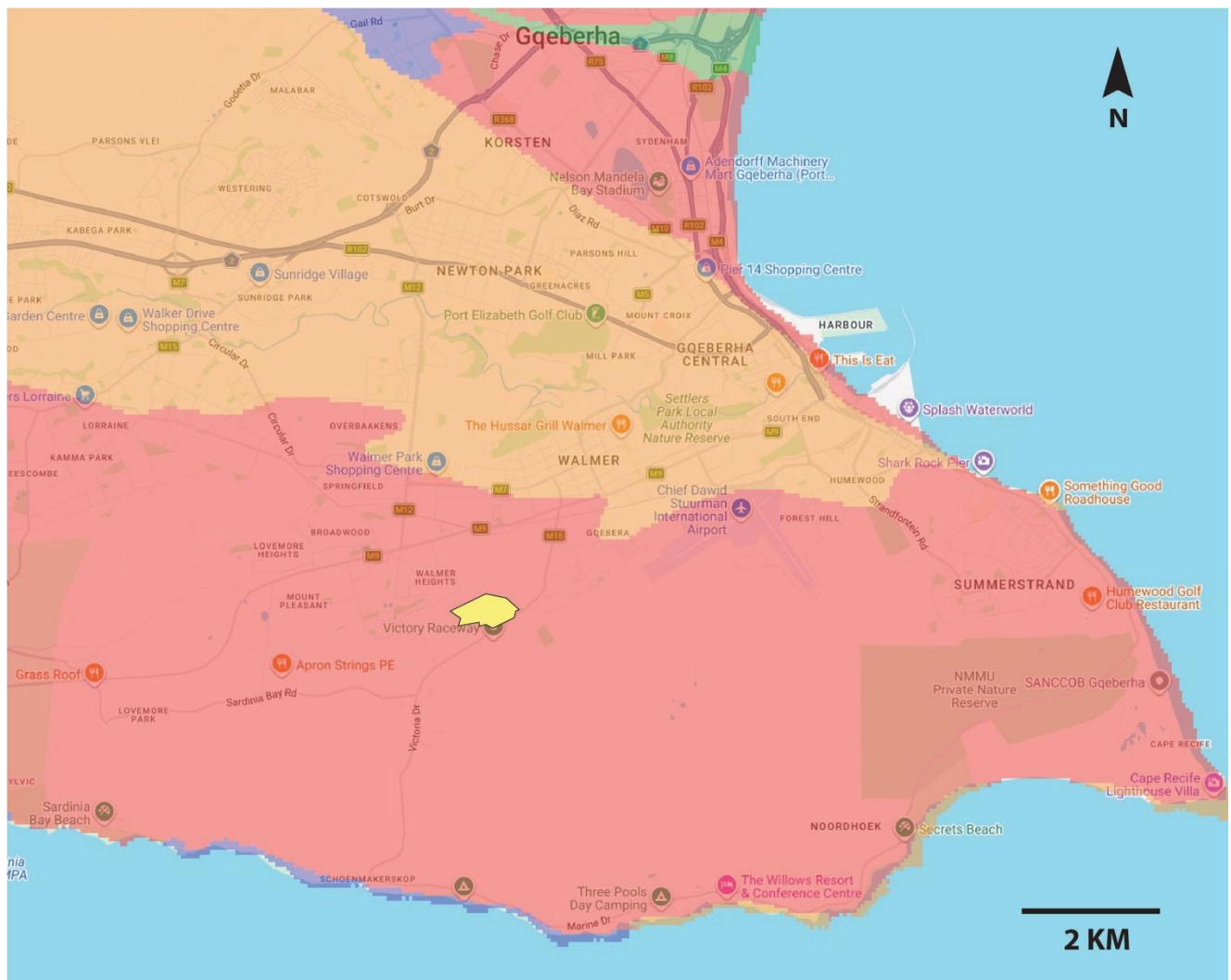
In addition to molluscan remains, other marine organisms such as echinoids, barnacles, and occasional crustacean fragments may also occur within these deposits. The preservation of shell beds and bioclastic concentrations reflects periods of high-energy coastal sedimentation, where wave action and shoreline processes concentrated skeletal material along beaches and shallow marine environments (Shone and Paszkowski, 1995). Such deposits commonly occur as coquinite layers or shell-rich sandstones within the stratigraphic succession of the Algoa Group. Vertebrate fossils are relatively rare within the Salnova Formation; however, isolated remains of marine vertebrates, including fish and occasionally marine mammals, have been reported from similar coastal Cenozoic deposits along the southern African margin (Roberts et al., 2006). The dynamic nature of coastal depositional systems often results in reworking and fragmentation of fossil material, which can affect preservation potential. Despite this, the fossil assemblages preserved within the Algoa Group remain important for understanding the development of coastal ecosystems and environmental changes along the southern African coastline during the Cenozoic.

From a palaeontological heritage perspective, the Algoa Group deposits are generally considered to have moderate fossil potential due to the occurrence of marine invertebrate assemblages and shell beds within calcareous coastal sediments. Where well preserved, these fossils can contribute to reconstructions of palaeoecology, sea-level fluctuations, and coastal environmental change during the late Cenozoic (Le Roux, 1989; Roberts et al., 2006).

3. PALAEONTOLOGICAL SENSITIVITY

The SAHRIS Palaeontological Sensitivity Map (Figure 4) classifies the rocks underlying the development area as very high, indicating that the palaeontological sensitivity of the area is considered significant (Almond and Pether, 2008). This classification reflects the recognised fossil potential of the geological units present within the study area and suggests that development activities may have the potential to impact palaeontological heritage resources. Despite the high palaeontological sensitivity assigned to these geological units, the study area itself is largely covered by superficial sediments and urban development, resulting in limited natural exposure of the underlying strata. The absence of visible outcrops reduces the likelihood of identifying fossiliferous horizons at the surface, although fossils may still occur within subsurface deposits that could be exposed during excavation activities.

Previous palaeontological work undertaken in the Walmer area provides additional context for assessing the likelihood of fossil occurrence. A specialist palaeontological assessment conducted for the proposed 132 kV powerline development in Walmer, Port Elizabeth, reported no fossil material within the investigated corridor and concluded that the development footprint was of low palaeontological significance due to the nature of the sediments and the limited exposure of fossil-bearing strata (Almond, 2016). This suggests that although the geological units of the Algoa Group may have fossil potential regionally, the probability of encountering significant fossil material within heavily developed and covered areas such as Walmer is relatively low. Nevertheless, because the underlying geological units belong to the Algoa Group and possess recognised fossil potential, the possibility of encountering isolated fossil material during excavation cannot be entirely excluded.



Colour	Sensitivity	Required Action
RED	VERY HIGH	field assessment and protocol for finds is required
ORANGE/YELLOW	HIGH	desktop study is required and based on the outcome of the desktop study, a field assessment is likely
GREEN	MODERATE	desktop study is required
BLUE	LOW	no palaeontological studies are required however a protocol for finds is required
GREY	INSIGNIFICANT/ZERO	no palaeontological studies are required
WHITE/CLEAR	UNKNOWN	these areas will require a minimum of a desktop study. As more information comes

		to light, SAHRA will continue to populate the map.
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Figure 3. Map and colour guide for palaeosensitivity of the study area. Development area marked with yellow polygon.

4. ASSUMPTIONS AND UNCERTAINTIES

The geological context of the study area, together with the currently documented palaeontological record, indicates that the underlying stratigraphy and lithological succession have the potential to preserve fossil material. The sedimentary sequence, which typically includes sandstones and associated calcareous coastal deposits with areas of unconsolidated overburden sands, forms part of geological units known to occur within zones of elevated palaeontological sensitivity. Fossil material may therefore be preserved within these sediments, particularly where fresh and unweathered strata are exposed during excavation activities. For this reason, the palaeontological potential of these lithologies should be considered during the planning and implementation of development activities.

By contrast, superficial Quaternary soils and recent surficial deposits are generally unlikely to contain fossil material of substantial scientific importance. Such unconsolidated sediments typically lack the conditions required for the long-term preservation of fossil remains and are therefore considered to have low palaeontological potential.

Uncertainties associated with this assessment primarily relate to subsurface conditions that cannot be directly observed prior to excavation. These include the thickness of soil cover, the depth of proposed excavations, and the extent to which intact sedimentary horizons may be intersected during construction. While natural surface exposures are limited, deeper excavations could potentially expose fossil-bearing strata. Consequently, although the overall likelihood of encountering fossil material within the development footprint is considered low to moderate, any fossils that are discovered may be of scientific significance and should be appropriately managed.

5. CONCLUSIONS AND RECOMMENDATIONS

The proposed Walmer Housing Development on ERF 11305 is located within Cenozoic coastal deposits of the Algoa Group, most likely belonging to the Salnova Formation. These deposits are known to contain marine fossil assemblages, particularly molluscs and other marine invertebrates preserved within calcareous sandstones and shell-rich horizons. As a result, the SAHRIS palaeosensitivity map classifies the geological units underlying the study area as having very high palaeontological sensitivity.

Despite this classification, the development area itself is situated within an urbanised environment where natural geological exposures are absent and the underlying strata are largely covered by soil and anthropogenic deposits. Previous palaeontological work undertaken in the broader Walmer area has reported no fossil material within similar development footprints and has indicated that the likelihood of encountering significant fossil resources within such settings is low.

Based on the results of this desktop assessment, the proposed development is therefore considered unlikely to result in significant impacts on palaeontological heritage resources. However, because the underlying geological units possess recognised fossil potential, the possibility of encountering isolated fossil material during excavation activities cannot be entirely ruled out.

It is therefore recommended that:

- The proposed development may proceed from a palaeontological perspective.
- No further specialist palaeontological studies are required prior to development.
- A Chance Find Protocol must be implemented during excavation activities.
- Should any fossil material be encountered during construction (e.g., fossil shells, bones, petrified wood, or trace fossils), work must cease in the immediate area and the Environmental

Control Officer (ECO) should notify a qualified palaeontologist and the relevant heritage authority so that the material can be appropriately recorded and managed.

Subject to the implementation of these measures, the proposed development is considered acceptable from a palaeontological heritage perspective.

While pre-excavation mitigation measures are not required, it is advisable to implement the following recommendations to ensure the discovery, preservation, and proper management of palaeontological resources during the proposed excavation and construction activities:

- **ECO Awareness:** The Environmental Compliance Officer (ECO) responsible for overseeing the project should be well-informed about the potential presence of fossils within the project area. It is essential that they are aware of the likelihood of uncovering fossils during excavation into the underlying strata.
- **Construction Manager Awareness:** The construction managers and supervisory personnel involved in the project should also be made aware of the possibility of encountering significant fossils on-site.
- **Exposure Inspection:** During excavation, it is crucial that any geological exposures encountered are thoroughly inspected for the presence of fossil remains, particularly fossils of high scientific value e.g. vertebrate teeth and bones. If any fossils of significance are discovered, the ECO should contact the appropriate authority to further investigate the area.
- **Qualified Palaeontologist Notification:** In the event of significant fossil discoveries, a qualified palaeontologist should be promptly notified. This palaeontologist should possess the expertise and qualifications necessary for accurately assessing and documenting the findings.
- **Recording and Sampling:** Any fossils of scientific value found within the excavation area should be regarded as of palaeontological interest. The palaeontologist should be

responsible for recording and sampling these fossils. The costs associated with this activity should be borne by the developer as part of their responsibility for preserving and managing palaeontological resources.

REFERENCES

- Almond, J. & Pether, J., 2008. Palaeontological heritage of the Western Cape. Heritage Western Cape Interim Technical Report. Unpublished report prepared for Heritage Western Cape.
- Almond, J.E., 2010. Palaeontological heritage assessment of the Coega IDZ, Eastern Cape Province. Natura Viva cc, Cape Town, 112 pp.
- Almond, J.E., 2016. Recommended exemption from further palaeontological studies: Proposed 132 kV powerline, Walmer, Port Elizabeth, Nelson Mandela Bay Municipality, Eastern Cape. Natura Viva cc Report.
- Hassan, S.M., Anderson, C.R. & de Wit, M., 2022. Facies analyses and depositional environments of the Mio–Pliocene Alexandria Formation, Algoa Basin, South Africa. *Journal of African Earth Sciences*, 192, 104555.
- Le Roux, F.G., 1989. The lithostratigraphy of Cenozoic deposits along the south-east Cape coast as related to sea-level changes. PhD thesis, Stellenbosch University, Stellenbosch.
- Le Roux, F.G., 1990. Palaeontological correlation of Cenozoic marine deposits of the southeastern, southern and western coasts, Cape Province. *South African Journal of Geology*, 93(3), pp.514–518.
- Le Roux, F.G., 2000. The geology of the Port Elizabeth–Uitenhage area. Explanation to Geological Map Sheets 3325DC, 3325DD, 3425BA Port Elizabeth and surrounding sheets. Council for Geoscience, Pretoria.
- Roberts, D.L., Botha, G.A., Maud, R.R. & Pether, J., 2006. Coastal Cenozoic deposits. In: Johnson, M.R., Anhaeusser, C.R. & Thomas, R.J. (eds.) *The Geology of South Africa*. Geological Society of South Africa and Council for Geoscience, Johannesburg & Pretoria, pp.605–628.
- Shone, R.W. & Paszkowski, T., 1995. A high-level terrace deposit of probable marine origin: The Vanstadens Conglomerate. *South African Journal of Geology*, 98(3), pp.256–262.
- Smuts, W., 1987. The Alexandria Formation in the Algoa Bay area. *South African Journal of Science*, 83(8), pp.504–505.

Appendix 1: Fossil Chance Finds Procedure

PROCEDURE: CHANCE FINDS OF PALAEONTOLOGICAL MATERIAL

Introduction

This document is aimed to inform workmen and foremen working on a construction and/or mining site. It describes the procedure to follow in instances of accidental discovery of palaeontological material (please see attached poster with descriptions of palaeontological material) during construction/mining activities. This protocol does not apply to resources already identified under an assessment undertaken under s. 38 of the National Heritage Resources Act (no 25 of 1999).

Fossils are rare and irreplaceable. Fossils tell us about the environmental conditions that existed in a specific geographical area millions of years ago. As heritage resources that inform us of the history of a place, fossils are public property that the State is required to manage and conserve on behalf of all the citizens of South Africa. Fossils are therefore protected by the National Heritage Resources Act and are the property of the State. Ideally, a qualified person should be responsible for the recovery of fossils noticed during construction/mining to ensure that all relevant contextual information is recorded.

Heritage Authorities often rely on workmen and foremen to report finds, and thereby contribute to our knowledge of South Africa's past and contribute to its conservation for future generations.

Training

Workmen and foremen need to be trained in the procedure to follow in instances of accidental discovery of fossil material, in a similar way to the Health and Safety protocol. A brief introduction to the process to follow in the event of possible accidental discovery of fossils should be conducted by the designated Environmental Control Officer (ECO) for the project, or the foreman or site agent in the absence of the ECO. It is recommended that copies of the attached poster and procedure are printed out and displayed at the site office so that workmen

may familiarise themselves with them and are thereby prepared in the event that accidental discovery of fossil material takes place.

Actions to be taken

One person in the staff must be identified and appointed as responsible for the implementation of the attached protocol in instances of accidental fossil discovery and must report to the ECO or site agent. If the ECO or site agent is not present on site, then the responsible person on site should follow the protocol correctly in order to not jeopardize the conservation and well-being of the fossil material.

Once a workman notices possible fossil material, he/she should report this to the ECO or site agent.

Procedure to follow if it is likely that the material identified is a fossil:

- i The ECO or site agent must ensure that all **work ceases** immediately in the vicinity of the area where the fossil or fossils have been found;
- ii The ECO or site agent must **inform HWC of the find immediately**. This information must include photographs of the findings and GPS co-ordinates;
- iii The ECO or site agent must compile a **Preliminary Report and fill in the *Fossil Discoveries: HWC Preliminary Record Form*** within 24 hours without removing the fossil from its original position. The **Preliminary Report** records basic information about the find including:
 - The date
 - A description of the discovery
 - A description of the fossil and its context (e.g. position and depth of find)
Where and how the find has been stored
 - Photographs to accompany the preliminary report (the more the better):
 - A scale must be used
 - Photos of location from several angles Photos of vertical

section should be provided

- Digital images of hole showing vertical section (side);
- Digital images of fossil or fossils.

- iv Upon receipt of this **Preliminary Report**, HWC will inform the ECO or site agent whether or not a rescue excavation or rescue collection by a palaeontologist is necessary.
- v **Exposed finds must be stabilized where they are unstable and the site capped, e.g. with a plastic sheet or sand bags.** This protection should allow for the later excavation of the finds with due scientific care and diligence. HWC can advise on the most appropriate method for stabilization.
- vi If the find cannot be stabilized, **the fossil may be collect with extreme care** by the ECO or the site agent and put aside and protected until HWC advises on further action. Finds collected in this way must be safely and securely stored in tissue paper and an appropriate box. Care must be taken to remove the all fossil material and any breakage of fossil material must be avoided at all costs.

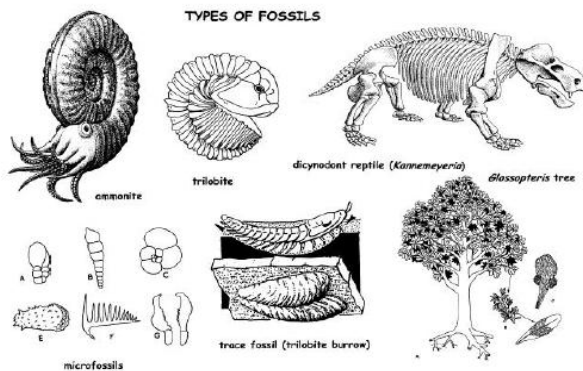
No work may continue in the vicinity of the find until HWC has indicated, in writing, that it is appropriate to proceed.

FOSSIL DISCOVERIES: HWC PRELIMINARY RECORDING FORM		
Name of project		
Name of fossil location		
Date of discovery		
Description of situation in which the fossil was found:		
Description of context in which the fossil was found:		
Description and condition of fossil identified:		
GPS coordinates:	Lat:	Long:
If no co-ordinates available then please describe the location:		
Time of discovery:		
Depth of find in hole:		
Photographs (tick as appropriate and indicate number of the photograph)	Digital image of vertical section (side)	
	Fossil from different angles	
	Wider context of the find	
Temporary storage (where it is located and how it is conserved)		
Person identifying the fossil	Name:	
	Contact:	
Recorder:	Name:	
	Contact:	
Photographer	Name:	
	Contact:	

Palaeontology: what is a fossil?

Fossils are the traces of ancient life (animal, plant or microbial) preserved within rocks and come in two forms:

- Body fossils preserve parts, casts or impressions of the original tissues of an organism (e.g. bones, teeth, wood, pollen grains); and
- Trace fossils such as trackways and burrows record ancient animal behaviour.



How to report chance fossil finds: What should I do if I find a fossil during construction/mining?

If you think you have identified a fossil:

Immediately inform the ECO or Site Agent.
He/she will then contact HWC and write a report
and if necessary operations will stop in that
specific area until the fossil is recovered

Heritage Western Cape
ceoheritage@westerncape.gov.za
021 483 5959
www.hwc.org.za

iLifa le...
Erfenis Wes-Kaap
Heritage Western Cape

Types of palaeontological finding - What does a fossil look like?

Fossils vary in size, from fossilised tree trunks and dinosaur bones down to very small animals or plants. Finds can be **individual fossils** (one isolated wood log or bone) or **clusters and beds** (several bones, teeth, animal or plant remains, trace fossils in close proximity or bones resembling part of a skeleton). A bed of fossils is a layer with many fossil remains.

Below there is a list of few examples of fossils which may be identified during excavations in the Western Cape.

Image	Description	Image	Description
	Leaves		Snail shells and other shells
	Fossil wood		Bones of larger animals
	The remains of fish and marine life (e.g. teeth, scales, starfish)		Large burrows made by moles and other animals
	Stromatolites		Traces made by burrowing insects (ants, wasps, dung-beetles etc.).
	Animal footprints		

Images provided by Dr John Almond

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